

Problems

Question 1:

Construct flowcharts for the following problems:

- (i) Print a currency conversion table for pounds, francs, marks, and lire to dollars.
- (ii) Find whether a given year is a leap year. Hint. A year is said to be a leap year if it is either divisible by 4 but not by 100 or divisible by 400.
- (iii) Validate a given year. Hints. The year in the date must be greater than zero, the months must lie between 1 and 12, and the days must lie between 1 and 31, depending on the month numbers.
- (iv) Show the time required by an advertising agency for its advertising program to run in Boston and on National Public Radio and to display the amount to be paid by the agency for its advertisement.
- (v) Calculate the commission of a salesman when sales and the region of the sales are given as input. The commission is calculated with the rules as follows:
 - a) No commission if sales < \$9,000 in Region A
 - b) 5.5% of sales < \$7,000 in Region B and when sales < \$13,000 in Region A
 - c) 7.5% of sales when sales > = \$14,000 in Region A and when sales > = \$13,000 in Region B.
- (vi) Accept three integers representing the angles of a triangle in degrees to determine whether they form a valid set of angles of a triangle. If it is not a valid set, then generate a message and terminate the process. If it is a valid set, then the process determines whether it is equiangular (all three angles are the same). It also determines if the triangle is right angled (has one angle with 90 degrees), obtuse angled (one angle above 90), or acute angled (all three angles are below 90 degrees). Finally, it shows conclusion about the triangle.
- (vii) Accept the lengths of the three sides of a triangle to validate whether they can be the sides of a triangle and then classify the triangle as equilateral (all three sides are equal), scalene (all three sides are different), or isosceles (exactly two sides are equal), and then to see whether it is a right angled triangle (the sum of the squares of two sides is equal to the square of the third side.) Hint. Three numbers are valid as the sides of a triangle if each one is positive and the sum of every two numbers is greater than the third.
- (viii) Allow the user to perform a simple task on a calculator on the basis of a given choice as follows: +, -, ×, /, or % representing the arithmetic operators A Average of two numbers X Maximum of two numbers M Minimum of two numbers S Square of two numbers Q Quit.
- (ix) An electricity board charges the following rates to domestic users to discourage large consumption of energy: for the first 100 units—\$.85 per unit for the next 200 units—\$1.45 per unit Beyond 300 units—\$1.85 per unit All users are charged a minimum of \$ 500.00. If the total cost is more than \$ 2,500.00, then an additional surcharge of 3% of the total cost is added to the total cost to determine the final bill.
- (x) To determine and print the minimum number of currency notes of the denominations: \$1, \$5, \$10, \$20, \$50, \$100, \$500 and \$1000 required to pay any given amount.

Question 2:

Write a program segment that inputs Age, displays "Yes, you can vote" if Age is 18 or older and displays "You are too young to vote" if Age is less than 18, and displays nothing else. Remember to use the comparison operator when checking the value of Age in your If-Then-Else statements.

Question 3:

Prompt for and input a saleswoman's sales for the month (in dollars) and her commission rate (as a percentage). Output her commission for that month. Note that you will need to convert the percentage to a decimal. You will need the following variables:

SalesAmount CommissionRate CommissionEarned

You will need the following formula:

$$\text{CommissionEarned} = \text{SalesAmount} * (\text{CommissionRate}/100)$$

Question 4:

Write a Raptor based program/flowchart to compute and display an investment's total interest and final value when the user inputs the original amount invested,

rate of following variables:

Interest (a Float)
Principal (a Float)
Rate (a Float)
FinalValue (a Float)
Time (an Integer)

You will need the following formulas:

$$\text{Interest} = \text{Principal} * \text{Rate} * \text{Time}$$

$$\text{FinalValue} = \text{Principal} + \text{Interest}$$

Question 5:

- a) Which control structure would be most appropriate for the following problems:
Sequential, Selection, Cascading Selection, or a Loop
- _____ Printing an appropriate message for a cadet's class year.
 - _____ Checking for a correct input and continually re-checking if incorrect.
 - _____ Computing the average GPA of your CS110 section.
 - _____ Determining the volume of a sphere given a radius.
 - _____ Initializing all of the variables at the beginning of a program.
 - _____ Determining whether a vowel, constant or digit has been typed in.
 - _____ Writing out a message if an integer variable contains a negative value.
 - _____ Writing "Odd" or "Even" depending on an integer variable's value.
 - _____ Writing out the squares of the numbers 1 through 100.
 - _____ Reading in scores until a user enters a negative number.
- b) Which of the following Decision expressions will always evaluate to "Yes", always evaluate to "No", or could possibly be either "Yes" or "No"?

_____ GR_Score > 100 or GR_Score < 90.
 _____ GR_Score > 100 and GR_Score < 90.
 _____ GR_Score < 100 or GR_Score > 90.
 _____ GR_Score < 100 and GR_Score > 90.

- c) Write a series of RAPTOR statements that determines if X has the value 1, 2, or 3, and prints out "ONE", "TWO", or "THREE" accordingly.

Question 6:

- a) Write a complete program that converts between degrees Fahrenheit and Celsius. The user must first enter the conversion that is desired (F to C or C to F) using any means you want and then enter the value to be converted. The formulas for conversion are:

$$F = 9/5 C + 32 \text{ and } C = 5/9 (F - 32)$$

- b) Write a complete program that plays the game of HI-LO. The program asks one user for a number between 1 and 100 and verifies that such a number has been entered. It then asks a second user for a guess and reads it in. If the guess is correct a congratulation message is written to the screen and the program ends. Otherwise the message "HI" or "LOW" is displayed (if the guess is higher or lower) and another guess is asked for and read in.

Question 7:

- a) Prompt for and input a saleswoman's sales for the month (in dollars) and her commission rate (as a percentage). Output her commission for that month.

Note that you will need to convert the percentage to a decimal. You will need the following variables:

SalesAmount CommissionRate CommissionEarned

You will need the following formula:

$$\text{CommissionEarned} = \text{SalesAmount} * (\text{CommissionRate}/100)$$

- b) The manager of the Super Supermarket would like to be able to compute the unit price for products sold there. To do this, the program should input the name and price of an item per pound and its weight in pounds and ounces. Then it should determine and display the unit price (the price per ounce) of that item and the total cost of the amount purchased. You will need the following variables:

ItemName Pounds Ounces
 PoundPrice TotalPrice UnitPrice

You will need the following formulas:

$$\text{UnitPrice} = \text{PoundPrice} / 16$$

$$\text{TotalPrice} = \text{PoundPrice} * (\text{Pounds} + \text{Ounces} / 16)$$

Question 8:

A used bicycle shop lends bicycles with the following rules: A deposit of \$150 must be made before taking any bicycle from the shop. The charges for hiring depend on the month in which it is hired. If the number of days of hire exceeds 15, a discount of 11% is offered. The hire-rate is determined as per the following rules:

Month name	Rate/day (in \$)
Jan. to Mar.	1.75
April to June	1.65
July to Sept.	1.50
Oct. to Dec.	1.15

In case of multiple day hiring, the rate is same for all days as on the first day of hiring. Develop a flowchart showing the logic to calculate the amount to be paid before taking any bicycle from the shop.

Task Analysis.

For the first three months of the year, that is, for the month numbers less than or equal to three, the rate of charge is the same. For the next three, but less than or equal to six, the rate of charge is the same. The next three months and the last three months are at a similar rate. To determine the charge for taking a bicycle from the shop, we require two inputs: the number of months in which a request is made for hire and the number of days for which the hire is effective. However, the month number given as input should be validated first and then the number of days for hire should be accepted as input. As the rates for the month numbers are given, the calculation of the charge of hiring after discount.

Question 9:

Draw a flowchart to show how to obtain the HCF and LCM of two numbers.

Task Analysis.

We know that HCF (Highest Common Factor) of two numbers is the largest number that can divide the two numbers without leaving any remainder and the LCM (Least Common Multiple) of two numbers is the smaller number that is divisible by both the numbers. The best way to obtain the HCF is the method of division in which one number is divided by another 70 ■ Flowchart and Algorithm Basics number to see if the remainder is zero. The divisor number is the HCF. If it is other than zero, then the divisor is made the dividend, the remainder is made the divisor, and the division is repeated to obtain the remainder again. This change of dividend and divisor is done repeatedly until we get the divisor that leaves zero as the remainder and the divisor in the last case is the HCF of the given numbers. The easiest way to find the LCM

is to use the relationship between the HCF and LCM. We know that the product of two numbers equals the product of their HCF and LCM. The LCM can be obtained by dividing the product of the given numbers by that HCF.

This is shown below:

$$\text{LCM} \times \text{HCF of two numbers} = \text{product of the two numbers}$$

$$\text{LCM of two numbers} = (\text{Product of the two numbers}) / (\text{HCF of the two numbers})$$

We require two inputs only: the two numbers we use to determine the HCF and LCM.

Question 10:

The charge for luggage on railways is calculated as shown below: For the first 40 kg. of weight, the charge is fixed at \$5.75. For every additional 18 kg. (or part thereof), the charge is calculated at the rate of \$ 3.88. If the total weight is less than 500 g, for weight beyond 500 kg., the charge is calculated at the rate of \$ 0.67 per kg. Develop a flowchart that will show the logic for calculating the charge of transporting a piece of luggage.

Task Analysis.

First, we check whether the weight is less than 40 kgs. If so, the determination of the charge is straightforward and needs no calculation. Next, we need to check whether it is less than 500 kgs. If it is under 500 kgs but more than 40 kgs, we must calculate the number of 18-kg intervals in the same way as we explained in the preceding solution. If the weight is greater than 500 kg, then we must calculate the excess weight beyond 500 kgs and number of intervals for the weights less than 500 (but greater than 40 kg).

Question 11:

Construct a flowchart for obtaining the sum of a given number of terms (N) for a given value of x in the following mathematical series:

$$X + X^2/2 + X^3/3 + \dots \text{ up to } N \text{ terms.}$$

Task Analysis.

We need to know the values of two unknown quantities as input: the value of X and that of N, the number of terms. We must find a general expression from which we can identify different terms. The expression X^l/l yields different terms for different values of l from 1 to N. We can evaluate the expression for some given value of X and the obtained value can be stored in some location, say S, that initially contains 0. Successive addition of the evaluated values for all the terms to the content of S will give us the desired sum.

Question 12:

Develop a flowchart to show how the sum of the following series can be obtained:

$$X - \frac{X^3}{3} + \frac{X^5}{5} - \frac{X^7}{7} + \frac{X^9}{9} \mp \dots \text{up to } n \text{ terms}$$

Task Analysis.

The inputs required are X and N. The general expression for any term I can be given by

$$t_i = \frac{X^{2I-1}}{2I-1} \text{ for } I = 1 \text{ to } N.$$

The flowchart will be the same as that for the next problem except that the statement.

$$S \leftarrow S + \frac{X^I}{I}$$

$$S \leftarrow S + X \frac{2I-1}{2I-1}$$

Question 13:

Develop a flowchart to show how to find out the sum of the following mathematical series:

$$X - \frac{1}{2}X^3 + \frac{1}{2} \cdot \frac{3}{4}X^5 - \frac{1}{2} \cdot \frac{3}{4} \cdot \frac{5}{6}X^7 + \dots \text{up to } n \text{ terms}$$

Task Analysis.

The inputs required here are X and N. The coefficients in this series bear a relationship with each other. The coefficient of the 2nd term is 1/2, that of the 3rd term is 1/2 x 3/4, and that of the 4th term is 1/2 x 3/4 x 5/6. We observe that each coefficient is a result of evaluating an expression of the form $\frac{2I-1}{2I}$ for $I = 1, 2, \dots$, an expression that generates an odd number in the numerator and an even number in the denominator. For $I = 1$, the coefficient is 1/2 if it is stored somewhere and it is multiplied by the value of the expression next time for $I = 2$, we get the 3rd coefficient; that is, we evaluate an expression of the form $C * \frac{(2I-1)}{2I}$ for $C = 1$ initially. The successive values of I from 1 to $n-1$ will give the coefficients from the 2nd to the nth term. Again, if we take $C = -C * \frac{(2I-1)}{2I}$, we get the coefficients with sign positive or negative. Now, multiplying the expression by X^{2I+1} for different values of I , we get the successive terms, except for the 1st term. Now, if we initialize S with the value of X and add the terms evaluated successively to the current value of S, we get the desired sum of the series.

Question 14:

The difference between two consecutive gas meter readings gives the amount of gas used in cubic feet. The gas charge is calculated using the following rules.

No. of terms consumed	Rate/term (in \$.)
≤ 75	1.05
> 75 but ≤ 150	1.25
> 150 but ≤ 250	1.50
> 250	2.25

The number of terms consumed is calculated by multiplying the amount of gas used by 1.06748. A meter rent of \$15 is also charged with each bill. The meters show 5-digit readings. Develop a program to print the gas bill for consumers.

Task Analysis.

The inputs required here are two consecutive meter readings. The difference between the readings will generally give us the amount of gas used, except in the extreme cases in which the previous reading becomes greater than the present reading. This extreme case occurs because the 5-digit meter resets when it reaches the reading of 100,000 (initially, the meter shows 00000 and it increases as the gas is consumed up to the largest value 99,999, after which one more unit of consumption will reset the meter because the meter does not possess the capacity to show 100,000). When the difference is obtained, it can be multiplied by 1.06748 to derive the number of terms consumed and then the calculation of the charge becomes straightforward.

Question 15:

Some three-digit numbers show the property that the sum of the factorials of the digits equals the numbers, for example, $145 = 1! + 4! + 5!$. Develop a flowchart to show how to determine all such numbers.

Task Analysis.

In this problem, too, we do not require any input from the terminal because we need to test all the three-digit numbers as in the preceding problem that can be generated serially. This solution requires a nested loop. The determination of the factorial was shown earlier. Here, the process should be repeated for each of the digits.

Question 16:

Determine the difference between two given dates. Construct a flowchart to show how to do it.

Task Analysis.

The inputs required are the day, month, and year number of two dates. The difference between the dates can then be computed by comparing the day numbers first and then by taking the difference; the month numbers are then compared and the number of months in between is then 106 ■ Flowchart and Algorithm Basics obtained. Now, when the two day numbers are compared, the day number of the smaller date may be larger than that of the greater date. To obtain the number of days in the difference, we add 30 to the smaller day number and subtract 1 from the corresponding month number. Otherwise, the number of days in the difference can be obtained through outright subtraction. To obtain the number of months in the difference, we compare the month number in the larger date with that in the smaller date. If the month number in the larger date is smaller than that of the smaller date, then 12 is added to the corresponding month number and then the number of months in the difference can be obtained through subtraction. One is subtracted from

the year number of the larger date. Otherwise, the number of months can be obtained through straightforward subtraction. There will be no such problem in obtaining the number of years because the year in the larger date will always be larger than the year in the smaller date.

Question 17:

Develop a flowchart to show how to find out all the groups of three successive numbers under 1000 that have the property that the square of the middle number is greater by unity than the product of the other two numbers (for example, $182 = 17 \times 19 + 1$).

Task Analysis.

We need to use numbers up to 1000. As any set of three successive numbers can satisfy the requirement, we may start checking from the first three natural numbers, i.e., 1, 2, and 3, and terminate the process when the largest of the numbers falls beyond 998.

Question 18:

Construct a flowchart to show how many number of elapsed days between two dates.

Task Analysis.

The problem is to determine the difference between two dates in the number of days. For example, the number of days between July 12, 1985, and July 26, 1985, is 14. But in the determination of number of days between two dates that differ widely, say, February 10, 1969 and July 21, 1988, the calculation is a bit more complex. Luckily, there is a formula that can be used to calculate the number of days between two dates. This is affected by computing the value of N for each of the two dates and then by taking the difference, where N is calculated as follows:

$$N = 1461 \times f(\text{year, month})/4 + 153 \times g(\text{month})/5 + \text{days}$$

where

$$f(\text{year, month}) = \text{year} - 1, \text{ if month} \leq 2 \\ = \text{year}, \text{ otherwise}$$

$$g(\text{month}) = \text{month} + 13, \text{ if month} \leq 2 \\ = \text{month} + 1, \text{ otherwise}$$

and all calculations are performed using integer arithmetic. As an example of applying the formula, to calculate the number of days between February 10, 1969 and July 21, 1988, we can calculate the values of N1 and N2 by substituting the appropriate values into the above formula as shown below.

$$\begin{aligned}
 N_1 &= 1461 \times f(1969, 2)/4 + 153 \times g(2)/5 + 10 \\
 &= (1461 \times 1968)/4 + 153 \times (2 + 13)/5 + 10 \\
 &= 2875248/4 + 2295/5 + 10 \\
 &= 718812 + 459 + 10 \\
 &= 719281 \\
 N_2 &= 1461 \times f(1988, 7)/4 + 153 \times g(7)/5 + 21 \\
 &= (1461 \times 1988)/4 + (153 \times 8)/5 + 21 \\
 &= 2904468/4 + 1224/5 + 21 \\
 &= 726117 + 244 + 21 \\
 &= 726382
 \end{aligned}$$

The number of elapsed days = $N_2 - N_1 = 719281 - 726382 = 7101$. The number of days between the two dates is 7101. The above formula is applicable for any date after March 1, 1900, (1 must be added to N for dates from March 1, 1800 to February 28, 1900, and 2 must be added for dates between March 1, 1700, and February 28, 1800, and so on).

Question 19:

Construct a flowchart to show how to determine the name of the starting day of any given year. Task Analysis. The name of a year can be determined based on a daycode that varies from 0 to 6. The day-code value 0 implies "Sunday," day-code value 1 implies "Monday," and so on. The day-code of a year X can be determined by the following formula:

$$\text{day-code} = \left[X + \left\lfloor \frac{x-1}{4} \right\rfloor - \left\lfloor \frac{x-1}{100} \right\rfloor + \left\lfloor \frac{x-1}{400} \right\rfloor \right] \bmod 7$$

where $[a]$ denotes the greatest integer less than or equal to a . For example $[5.6]$ is 5. This is also known as floor value. Mod 7 implies remainder after division by 7. The day-code value can also be computed by taking the value of N computed in the preceding problem and then subtracting 621,049 from it and then taking the result modulo 7. Here we use the previous procedure to construct the following flowchart because that is true for any year.

Question 20:

Construct flowcharts to show the following:

- (i) Print multiplication tables from 1 to 5
- (ii) Sum the digits of a given number until it is reduced to a single digit
- (iii) Create a pyramid of numbers consisting of a given number of lines. For example, if the given number is 5, then we should see the following:

```

      1
    1 2 1
  1 2 3 2 1
1 2 3 4 3 2 1
1 2 3 4 5 4 3 2 1

```

- (iv) Sum the digits of a given number.
- (v) A menu of fruits as given below that accepts the user's option. Calculate the cost of fruits and repeat the same until the user's option is exited. Display the cost of each item and the total amount to be paid by the customer.

Fruits Menu		
	Fruits	Cost per Pound. (in \$)
1.	Mango	5.00
2.	Apple	3.00
3.	Grapes	2.00
4.	Exit	

- (vi) The following patterns with flexible dimensions as supplied by the uses:

(a)

```

      ★
    ★ ★
  ★ ★ ★
★ ★ ★ ★

```

(b)

```

      ★
    ★ ★ ★
  ★ ★ ★ ★
★ ★ ★ ★ ★

```

(c)

```

★ ★ ★ ★ ★ ★ ★
★ ★ ★ ★ ★ ★ ★
★ ★ ★ ★ ★ ★ ★
    
```

(d)

```

★ ★ ★ ★ ★ ★ ★ ★
★
★ ★ ★ ★ ★ ★ ★ ★
    
```

(e)

```

★ ★ ★ ★ ★
★ ★ ★ ★
★ ★ ★
★ ★
★
    
```

(f)

```

          ★
        ★ ★ ★
      ★ ★ ★ ★ ★
    ★ ★ ★ ★ ★ ★ ★
  ★ ★ ★ ★ ★ ★ ★
    ★ ★ ★
      ★
    
```

- (vii) Read a six-digit positive integer. If the number is even, add up its digits. Otherwise, multiply the individual digits and print the result.
 - (viii) Obtain the decimal equivalent of a binary number.
 - (ix) Display all characters represented by the ASCII numbers from 25 to 100 (x) Determine the value of an exponential expression of the form a^x , where a is any number and x is any integer
 - (x) Determine the HCF of n given numbers (xii) Determine the maximum and the minimum ones of n given numbers
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- (xi) Determine all the permutations of the numbers less than or equal to some given number n . For example, if $n = 123$, then the permutations are:

```

123
321
231
132
213
312
    
```